ALLIANCE UNIVERSITY • SCHOOL OF COMPUTING • DEPARTMENT OF COMPUTER APPLICATIONS

Assignment 7: Sample Data Insertion & Population Audit

Realistic Enterprise Dataset Ingestion for Smart Library Management System

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Program / Semester: BCA • Semester III
Course: Database Management Systems (DBMS Lab)
Deliverable: Individual Assignment 7 (PDF)

1. Introduction & Population Strategy

A database design cannot be adequately validated without populating it with realistic, diverse sample data that accurately mirrors actual institutional operations. For the **Smart Library Management System (SLMS)**, data insertion must reflect the real-world operational workflows of an academic university library—including prestigious university presses, canonical computer science textbooks, multi-author collaborations, diverse patron categories (undergraduate students, research faculty, alumni, and external guests), multi-tiered staff hierarchies, and complex circulation lifecycles (active loans, on-time returns, overdue checkouts, and penalty assessments).

This document details the complete SQL Data Manipulation Language (DML) insertion scripts executed across all 9 tables, explains the relational strategy employed to preserve foreign key order, and presents full tabular verification dumps confirming 100% compliance with all defined constraints.

2. Sequential Ingestion Scripts (SQL DML)

Data insertion follows strict topological sorting based on relational dependencies: lookup dimensions are populated first, followed by catalog titles and junction records, and concluded with circulation events and fine assessments.

2.1 Dimension Tables: Publishers, Authors, and Categories

```
-- 1. Insert Academic Publishers
INSERT INTO publishers (publisher_id, publisher_name, country, contact_email) VALUES
(1, 'MIT Press', 'United States', 'info@mitpress.mit.edu'),
(2, 'O''Reilly Media', 'United States', 'support@oreilly.com'),
(3, 'Pearson Education', 'United Kingdom', 'contact@pearson.com'),
(4, 'Springer Nature', 'Germany', 'service@springernature.com'),
(5, 'Oxford University Press', 'United Kingdom', 'enquiry@oup.com');

-- 2. Insert Canonical Authors
INSERT INTO authors (author_id, author_name, nationality) VALUES
(1, 'Donald E. Knuth', 'American'),
(2, 'Stuart Russell', 'British'),
(3, 'Peter Norvig', 'American'),
(4, 'Martin Kleppmann', 'British'),
(5, 'Abraham Silberschatz', 'American'),
(6, 'Henry F. Korth', 'American'),
(7, 'Robert C. Martin', 'American'),
(8, 'Thomas H. Cormen', 'American');

-- 3. Insert Subject Categories
INSERT INTO categories (category_id, category_name, description) VALUES
(1, 'Computer Science & AI', 'Foundational algorithms, data structures, and artificial
intelligence'),
(2, 'Database Systems', 'Relational design, indexing, transaction processing, and NoSQL
architecture'),
(3, 'Software Engineering', 'Software design patterns, clean architecture, refactoring, and QA'),
(4, 'Distributed Systems', 'Cloud computing, data-intensive architectures, and distributed storage'),
(5, 'Mathematics & Logic', 'Discrete mathematics, probability, mathematical reasoning, and
optimization'),
(6, 'Information Systems', 'Enterprise systems, digital governance, and technology management');
```

2.2 Catalog Titles & Authorship Junction (Books & Book_Authors)

```
-- 4. Insert Catalog Books
INSERT INTO books (book_id, isbn, title, category_id, publisher_id, publication_year, total_copies,
available_copies, price) VALUES
(1, '978-0262033848', 'Introduction to Algorithms (4th Ed)', 1, 1, 2022, 8, 5, 89.99),
(2, '978-0136042594', 'Artificial Intelligence: A Modern Approach', 1, 3, 2020, 6, 2, 99.50),
(3, '978-1449373320', 'Designing Data-Intensive Applications', 4, 2, 2017, 10, 6, 45.00),
(4, '978-0078022159', 'Database System Concepts (7th Ed)', 2, 4, 2019, 7, 3, 85.00),
(5, '978-0132350884', 'Clean Code: A Handbook of Agile Software Craftsmanship', 3, 3, 2008, 9, 7,
42.50),
(6, '978-0201896831', 'The Art of Computer Programming, Vol 1', 1, 3, 1997, 4, 4, 75.00),
(7, '978-1491950357', 'Designing Distributed Systems', 4, 2, 2018, 5, 3, 39.99),
(8, '978-0134494166', 'Clean Architecture: A Craftsman''s Guide', 3, 3, 2017, 6, 4, 44.00),
(9, '978-0199572830', 'Information: A Very Short Introduction', 6, 5, 2010, 5, 5, 11.95),
(10, '978-0262534802', 'Deep Learning (Adaptive Computation & Machine Learning)', 1, 1, 2016, 7, 2,
80.00);

-- 5. Insert Book Authorship Links (Resolving M:N Relationships)
INSERT INTO book_authors (book_id, author_id) VALUES
(1, 8), -- Cormen -> Intro to Algorithms
(2, 2), (2, 3), -- Russell & Norvig -> AI A Modern Approach
(3, 4), -- Kleppmann -> DDIA
(4, 5), (4, 6), -- Silberschatz & Korth -> Database System Concepts
(5, 7), -- Martin -> Clean Code
(6, 1), -- Knuth -> Art of Computer Programming
(7, 4), -- Kleppmann -> Designing Distributed Systems
(8, 7), -- Martin -> Clean Architecture
(10, 2); -- Russell -> Deep Learning reference
```

2.3 Patrons and Staff Hierarchies (Members & Staff)

```
-- 6. Insert Library Patrons
INSERT INTO members (member_id, full_name, email, phone, membership_type, join_date, status,
max_books_allowed) VALUES
(1, 'Sanjeev Adarsh', 'sanjeev.a@alliance.edu.in', '+91-9876543210', 'Student', '2025-08-01',
'Active', 5),
(2, 'Hemanth Krishna', 'hemanth.k@alliance.edu.in', '+91-9876543211', 'Student', '2025-08-05',
'Active', 5),
(3, 'Vigneshvar K.K.', 'vigneshvar.k@alliance.edu.in', '+91-9876543212', 'Student', '2025-08-10',
'Active', 5),
(4, 'Mahesh M.', 'mahesh.m@alliance.edu.in', '+91-9876543213', 'Student', '2025-08-12', 'Active', 5),
(5, 'Dr. Arisudan Sharma', 'asharma@alliance.edu.in', '+91-9876543214', 'Faculty', '2024-01-15',
'Active', 10),
(6, 'Prof. Renita Joy', 'renita.joy@alliance.edu.in', '+91-9876543215', 'Faculty', '2024-06-20',
'Active', 10),
(7, 'Dhananjay Shankar', 'dhananjay.s@alumni.alliance.edu.in', '+91-9876543216', 'Alumni', '2023-05-
18', 'Active', 3),
(8, 'Kabila Varshini', 'kabila.v@external.org', '+91-9876543217', 'Guest', '2026-02-01', 'Suspended',
2);

-- 7. Insert Staff Members
INSERT INTO staff (staff_id, staff_name, role, email, hire_date) VALUES
(1, 'David Miller', 'Head Librarian', 'dmiller@smartlibrary.org', '2021-03-01'),
(2, 'Sunita Rao', 'Assistant Librarian', 'srao@smartlibrary.org', '2022-07-15'),
(3, 'Praveen Nair', 'Cataloger', 'pnair@smartlibrary.org', '2023-02-10'),
(4, 'Anita Desai', 'Circulation Clerk', 'adesai@smartlibrary.org', '2024-09-01');
```

2.4 Operational Transactions: Loans and Fines

```
-- 8. Insert Circulation Loan Records (Covering Active, Returned, and Overdue states)
INSERT INTO loans (loan_id, book_id, member_id, staff_id, issue_date, due_date, return_date, status)
VALUES
(1, 1, 1, 4, '2026-09-01', '2026-09-15', '2026-09-14', 'Returned'), -- On-time return
(2, 2, 1, 4, '2026-09-15', '2026-09-29', '2026-10-05', 'Returned'), -- 6 days late ($12 fine)
(3, 3, 2, 2, '2026-09-10', '2026-09-24', '2026-09-22', 'Returned'), -- On-time return
(4, 4, 3, 4, '2026-09-20', '2026-10-04', '2026-10-12', 'Returned'), -- 8 days late ($16 fine)
(5, 5, 4, 4, '2026-09-25', '2026-10-09', '2026-10-09', 'Returned'), -- On-time return
(6, 2, 5, 1, '2026-10-01', '2026-10-31', NULL, 'Issued'),           -- Active faculty loan
(7, 3, 6, 2, '2026-10-05', '2026-11-05', NULL, 'Issued'),           -- Active faculty loan
(8, 1, 2, 4, '2026-10-01', '2026-10-15', NULL, 'Overdue'),          -- Overdue (8 days overdue)
(9, 4, 5, 1, '2026-10-08', '2026-11-08', NULL, 'Issued'),           -- Active faculty loan
(10, 10, 1, 4, '2026-10-02', '2026-10-16', NULL, 'Overdue'),         -- Overdue (7 days overdue)
(11, 7, 3, 2, '2026-10-10', '2026-10-24', NULL, 'Issued'),          -- Active student loan
(12, 10, 4, 4, '2026-10-12', '2026-10-26', NULL, 'Issued');         -- Active student loan

-- 9. Insert Fine Penalty Assessments
INSERT INTO fines (fine_id, loan_id, fine_amount, paid_amount, payment_status, payment_date) VALUES
(1, 2, 12.00, 12.00, 'Paid', '2026-10-05'),
(2, 4, 16.00, 16.00, 'Paid', '2026-10-12'),
(3, 8, 16.00, 0.00, 'Unpaid', NULL),
(4, 10, 14.00, 7.00, 'Partially Paid', '2026-10-20');
```

3. Comprehensive Ingestion Audit & Data Verification

The database engine confirmed zero constraint violations during ingestion. The tables below present verified catalog dumps across core entities:

3.1 Catalog Holdings Verification (`books` Table)

ID	ISBN	Title	Cat ID	Pub ID	Year	Total	Avail	Price
1	978-0262033848	Introduction to Algorithms (4th Ed)	1	1	2022	8	5	$89.99
2	978-0136042594	Artificial Intelligence: A Modern Approach	1	3	2020	6	2	$99.50
3	978-1449373320	Designing Data-Intensive Applications	4	2	2017	10	6	$45.00
4	978-0078022159	Database System Concepts (7th Ed)	2	4	2019	7	3	$85.00
5	978-0132350884	Clean Code: Agile Software Craftsmanship	3	3	2008	9	7	$42.50
6	978-0201896831	The Art of Computer Programming, Vol 1	1	3	1997	4	4	$75.00
7	978-1491950357	Designing Distributed Systems	4	2	2018	5	3	$39.99
8	978-0134494166	Clean Architecture: A Craftsman's Guide	3	3	2017	6	4	$44.00
9	978-0199572830	Information: A Very Short Introduction	6	5	2010	5	5	$11.95
10	978-0262534802	Deep Learning (Adaptive Computation)	1	1	2016	7	2	$80.00

3.2 Circulation Loan Audit (`loans` Table)

Loan ID	Book Title (Resolved via FK)	Member Name (Resolved via FK)	Staff Issuer	Issue Date	Due Date	Return Date	Status
1	Introduction to Algorithms (4th Ed)	Sanjeev Adarsh	Anita Desai	2026-09-01	2026-09-15	2026-09-14	Returned
2	Artificial Intelligence: A Modern Approach	Sanjeev Adarsh	Anita Desai	2026-09-15	2026-09-29	2026-10-05	Returned
3	Designing Data-Intensive Applications	Hemanth Krishna	Sunita Rao	2026-09-10	2026-09-24	2026-09-22	Returned
4	Database System Concepts (7th Ed)	Vigneshvar K.K.	Anita Desai	2026-09-20	2026-10-04	2026-10-12	Returned
5	Clean Code	Mahesh M.	Anita Desai	2026-09-25	2026-10-09	2026-10-09	Returned
6	Artificial Intelligence: A Modern Approach	Dr. Arisudan Sharma	David Miller	2026-10-01	2026-10-31	NULL	Issued
7	Designing Data-Intensive Applications	Prof. Renita Joy	Sunita Rao	2026-10-05	2026-11-05	NULL	Issued
8	Introduction to Algorithms (4th Ed)	Hemanth Krishna	Anita Desai	2026-10-01	2026-10-15	NULL	Overdue
9	Database System Concepts (7th Ed)	Dr. Arisudan Sharma	David Miller	2026-10-08	2026-11-08	NULL	Issued
10	Deep Learning	Sanjeev Adarsh	Anita Desai	2026-10-02	2026-10-16	NULL	Overdue

Loan ID	Book Title (Resolved via FK)	Member Name (Resolved via FK)	Staff Issuer	Issue Date	Due Date	Return Date	Status
11	Designing Distributed Systems	Vigneshvar K.K.	Sunita Rao	2026-10-10	2026-10-24	NULL	Issued
12	Deep Learning	Mahesh M.	Anita Desai	2026-10-12	2026-10-26	NULL	Issued

3.3 Fine Ledger Verification (`fines` Table)

Fine ID	Loan Reference	Fine Assessed	Amount Paid	Outstanding Balance	Payment Status	Settlement Date
1	Loan #2 (Sanjeev Adarsh)	$12.00	$12.00	$0.00	Paid	2026-10-05
2	Loan #4 (Vigneshvar K.K.)	$16.00	$16.00	$0.00	Paid	2026-10-12
3	Loan #8 (Hemanth Krishna)	$16.00	$0.00	$16.00	Unpaid	Pending
4	Loan #10 (Sanjeev Adarsh)	$14.00	$7.00	$7.00	Partially Paid	2026-10-20

> ***Population Summary:*** *The dataset populates all 9 relations with sufficient operational density. All primary keys, unique constraints, check ranges, and parent-child foreign key relationships were verified without warnings. The populated database provides a complete real-world foundation for executing the analytical SQL queries and multi-table joins evaluated in Assignments 8 and 9.*